

**PACKAGE INSERT
DEXMETETOMIDINE KABI**

0.1 mg/mL Concentrate for Infusion
Dexmedetomidine Hydrochloride Injection

Product Name

Dexmedetomidine Kabi

0.1 mg/mL Concentrate for Infusion

Name and Strength of Active Ingredient (s)

Dexmedetomidine as Dexmedetomidine Hydrochloride USP 0.1 mg/mL

Product Description

Dexmedetomidine Kabi 0.1 mg/mL is sterile solution for Infusion.

Dexmedetomidine Kabi 0.1 mg/mL is a clear, colourless solution free from visible extraneous matter.

Dexmedetomidine Hydrochloride has been shown to be compatible when administered with the following diluents/ intravenous fluids: 0.9% Sodium Chloride solution, Lactated Ringers solution, 5% Dextrose in Water, 20% Mannitol, 100 mg/mL Magnesium sulfate solution and 0.3% Potassium chloride solution. After dilution, the product is a clear, colourless solution free from visible extraneous matter.

The diluents/ intravenous fluids are not being supplied together with this product.

Dexmedetomidine Kabi 0.1 mg/mL is filled in 2 mL USP Type I, clear glass vial administered by I.V. infusion use.

Excipients:

Sodium Chloride USP

Water for Injection USP

Pharmacodynamics

Pharmacotherapeutic group

Psycholeptics, other hypnotics and sedatives.

Mechanism of action and Pharmacodynamic effects:

Dexmedetomidine is a selective α_2 -adreno-receptor agonist with sedative and a broad range of pharmacological properties. α_2 - Selectivity is observed in animals following slow intravenous infusion of low and medium doses (10-300 mcg / kg). Both α_1 and α_2 activities are observed following slow intravenous infusion of high doses (≥ 1000 mcg/kg) or with rapid intravenous administration.

Pharmacokinetics

Following intravenous administration, Dexmedetomidine exhibits the following pharmacokinetic characteristics: rapid distribution phase with a distribution half-life ($t_{1/2\alpha}$) of about six minutes; and a terminal elimination half-life ($t_{1/2\beta}$) approximately two hours. Dexmedetomidine exhibits linear kinetics in the dosage range of 0.2 to 0.7 mcg/kg/hr when administered by IV infusion for up to 24 hours.

Distribution:

The steady-state volume of distribution (Vss) of Dexmedetomidine is approximately 118 liters. The average protein binding is 94%, and is significantly decreased in subjects with hepatic impairment.

Metabolism

Dexmedetomidine undergoes almost complete biotransformation with very little unchanged Dexmedetomidine excreted in urine and faeces. Biotransformation involves both direct glucuronidation as well as cytochrome P450 mediated metabolism.

The major metabolic pathways of Dexmedetomidine are: direct N- glucuronidation to inactive metabolites; aliphatic hydroxylation (mediated primarily by CYP2A6) of Dexmedetomidine to generate 3-hydroxy Dexmedetomidine, the glucuronide of 3-hydroxy Dexmedetomidine, and 3-carboxy Dexmedetomidine, and N- methylation of Dexmedetomidine to generate 3-hydroxy N-methyl Dexmedetomidine, 3-carboxy N-methyl Dexmedetomidine and N-methyl O-glucuronide Dexmedetomidine.

Elimination:

The terminal elimination half-life ($t_{1/2}$) of Dexmedetomidine is approximately 2 hours and clearance is estimated to be approximately 39 L/h.

Hepatic Function Impairment:

In subjects with varying degrees of hepatic impairment, clearance values are lower than in healthy subjects. It is necessary to reduce the dose depending on the degree of hepatic impairment.

Renal Function Impairment:

Dexmedetomidine hydrochloride pharmacokinetics (Cmax, Tmax, AUC, $t_{1/2}$, CL and Vss) do not vary in subjects with severe renal impairment (Cr CL: <30 mL/min). Since the majority of metabolites are excreted in the urine, it is possible that the metabolites may accumulate upon long-term infusions in patients with impaired renal function.

Elderly:

The pharmacokinetic profile of Dexmedetomidine hydrochloride was not altered by age, however as with many drugs, the elderly may be more sensitive to the effects of Dexmedetomidine.

Children:

The pharmacokinetic profile of Dexmedetomidine hydrochloride has not been studied in children.

Indications

Intensive Care Unit Sedation:

Dexmedetomidine hydrochloride is indicated for sedation of initially intubated and mechanically ventilated patients during treatment in an intensive care setting. Dexmedetomidine hydrochloride should be administered by continuous infusion not to exceed 24 hours.

Dexmedetomidine hydrochloride has been continuously infused in mechanically ventilated patients prior to extubation, during extubation, and post-extubation. It is not necessary to discontinue Dexmedetomidine hydrochloride prior to extubation.

Procedural Sedation:

Dexmedetomidine hydrochloride is indicated for sedation of non-intubated patients prior to and/or during surgical and other procedures.

Recommended Dose and Mode of administration

Dexmedetomidine should be administered using a controlled infusion device. Add 2 mL of the preparation to 48 mL of sterile 0.9% sodium chloride solution to obtain a total of 50 mL solution with a concentration of 4 mcg/mL of Dexmedetomidine. This solution is to be used for both initiation and maintenance of sedation. Dose reduction is necessary in patients who are elderly or with impaired hepatic or renal function.

Patients receiving Dexmedetomidine have been observed to be rousable and alert when stimulated. This is an expected component of Dexmedetomidine sedation and should not be considered a lack of efficacy in the absence of other clinical signs and symptoms.

Adults

Initiation of Intensive care unit sedation:

Dose of Dexmedetomidine hydrochloride should be individualized and titrated to the desired clinical effect. For adult patients, it is recommended to initiate Dexmedetomidine hydrochloride with a loading dose of 1.0 mcg/kg over 10 minutes, followed by a maintenance infusion in the range of 0.2 to 0.7 mcg/kg/hr. The rate of the maintenance infusion can be adjusted in order to achieve the desired level of sedation. As a guide, it is recommended that 0.4 mcg/kg/hr should be the initial maintenance infusion. The dose should be titrated against the response of the patient. If after approximately 5 minutes sedation is not adequate, the rate of infusion can be increased in increments of 0.1 mcg/kg/hr or higher.

Dosage for initiation of procedural sedation:

For adult patients: a loading infusion of 1 mg/kg over 10 minutes. For less invasive procedures, a loading infusion of 0.5 mcg/kg given over 10 minutes may be suitable.

For patients over 65 years of age: a loading infusion of 0.5 mcg/kg over 10 minutes.

Dosage for maintenance of procedural sedation:

For adult patients; the maintenance infusion is generally initiated at 0.6 mcg/kg/hr and titrated to achieve desired clinical effect with doses ranging from 0.2 to 1 mcg/kg/hr.

For a wake fiberoptic intubation patient; a maintenance infusion of 0.7 mcg/kg/hr is recommended until the endotracheal tube is secured.

Impaired Hepatic Function

Dosage reduction is needed for patients with hepatic impairment, as Dexmedetomidine hydrochloride is metabolized primarily in the liver.

Impaired Renal Function

Dosage reduction is needed for patients with renal impairment, as the majority of metabolites are excreted in the urine.

Elderly

Dexmedetomidine should be titrated against the response of the patient. Elderly patients (over 65 years) often require lower doses of Dexmedetomidine.

Children

Safety and efficacy of Dexmedetomidine Hydrochloride has not been studied in children.

Preparation of infusion

To prepare the infusion, withdraw 2 mL of Dexmedetomidine Hydrochloride concentrate for solution for infusion and add to 48 mL of 0.9% sodium chloride to total 50 mL. Shake gently to mix well. A controlled infusion device should be used to administer Dexmedetomidine hydrochloride.

Compatibility

Dexmedetomidine Hydrochloride has been shown to be compatible when administered with the following intravenous fluids: Lactated Ringers, 5% dextrose in Water, 0.9% Sodium Chloride in Water, 20% Mannitol in Water, Lactated Ringer's solution, 100 mg/mL magnesium sulfate solution, 0.3% potassium chloride solution. Dexmedetomidine has been found to be compatible with water solutions of the following drugs when administered via Y-site injection: thiopental sodium, Vecuronium Bromide, Pancuronium bromide, glycopyrrolate bromide, Phenylephrine hydrochloride.

Incompatibilities

Compatibility of Dexmedetomidine Hydrochloride with co-administration of blood, serum, or plasma has not been established. Dexmedetomidine must not be mixed with other medicinal products or diluents except those mentioned above.

Contraindications

- Hypersensitivity to the active substance or to any of the excipients in the formulation.
- Advanced heart block (grade 2 or 3) unless paced.
- Uncontrolled hypotension.
- Acute cerebrovascular conditions.

Warnings and Precautions

Continuous electrocardiogram (ECG), blood pressure and oxygen saturation monitoring are recommended during infusion of Dexmedetomidine. Caution should be exercised in patients with preexisting severe bradycardia disorders (i.e. Advanced heart block), or patients with preexisting severe ventricular dysfunction (eg. ejection fraction <30%) including congestive heart failure and cardiac failure in whom sympathetic tone is critical for maintaining haemodynamic balance.

Dexmedetomidine decreases sympathetic nervous activity and therefore, these effects may be expected to be most pronounced in patients with desensitised autonomic nervous system control.

Prevention of hypotension and bradycardia should take into consideration the hemodynamic stability of the patient and normovolemia prior to administration of Dexmedetomidine.

Patients who are hypovolemic may become hypotensive under Dexmedetomidine therapy. Therefore, fluid supplementation should be administered prior to and during the administration of Dexmedetomidine.

Based on clinical experience with Dexmedetomidine, if medical intervention is required, treatment may include decreasing or stopping the infusion of Dexmedetomidine, increasing the rate of IV fluid administration, elevation of the lower extremities, and use of pressor agents.

Elderly patients over 65 years of age or diabetic patients are more prone to hypotension with the administration of Dexmedetomidine.

Transient hypertension has been observed primarily during the loading dose infusion associated with initial peripheral vasoconstrictive effects of Dexmedetomidine and relatively higher plasma concentrations achieved during the loading infusion. In such cases reduction of the infusion rate may be considered. Following the loading dose infusion, the central effects of Dexmedetomidine dominate and the blood pressure usually decreases. Clinical events of bradycardia and sinus arrest have been associated with Dexmedetomidine administration in young, healthy volunteers with high vagal tone or with different routes of administration including rapid intravenous or bolus administration of Dexmedetomidine. In clinical trials, glycopyrrolate or atropine was effective in the treatment of most episodes of Dexmedetomidine induced bradycardia. However, in some patients with significant cardiovascular dysfunction, more advanced resuscitative measures were required.

Dexmedetomidine may cause reduced lacrimation. Lubrication of the patient's eyes should be considered when administering Dexmedetomidine to avoid corneal dryness.

Parental drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

Withdrawal

Although not specifically studied, if Dexmedetomidine is administered chronically (for >24 hours) and stopped abruptly, withdrawal symptoms similar to those reported for another α_2 -adrenergic agent, clonidine, may result. These symptoms include nervousness, agitation and headache, accompanied or followed by a rapid rise in blood pressure and elevated catecholamine concentrations in the plasma. Dexmedetomidine should not be administered for greater than 24 hours.

Interactions with other medicaments

General: In vitro studies indicate the clinically relevant cytochrome P450 mediated drug interactions are unlikely.

Anesthetics/Sedatives/Hypnotics/Opioids:

Co-administration of Dexmedetomidine hydrochloride is likely to lead to an enhancement of effects with anesthetics, sedatives, hypnotics and opioids. Due to Pharmacodynamic effects, when co-administered with Dexmedetomidine hydrochloride, a reduction in dosage with these agents may be required. Additionally, in situations where other vasodilators or negative chronotropic agents are administered, co-administration of Dexmedetomidine could have an additive Pharmacodynamic effect and should be administered with caution and careful titration.

Clinical events of bradycardia or hypotension may be potentiated when Dexmedetomidine is used concurrently with propofol or Midazolam. Therefore, consider a reduction in the dose of

Midazolam or propofol.

Pregnancy and Lactation

Pregnancy:

There are no adequate and well-controlled studies in pregnant women. Dexmedetomidine hydrochloride should be used during pregnancy only if the potential benefits justify the potential risk to the foetus.

Lactation:

It is not known whether Dexmedetomidine Hydrochloride is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when Dexmedetomidine hydrochloride infusion is administered to a nursing woman.

Undesirable Effects

Overall, the most frequently observed adverse events included hypotension, hypertension, bradycardia, dry mouth and nausea.

Other reported adverse reactions include, amongst others, fever, pain, rigors, blood pressure fluctuations, arrhythmia, AV Block, cardiac arrest, extrasystoles, atrial fibrillation, T-wave inversion, tachycardia, oedema, cyanosis, angina pectoris, myocardial ischaemia, pulmonary oedema, dizziness, headache, neuralgia, neuritis, speech disorder, abdominal pain, diarrhoea, vomiting, ascites, eructation, acidosis, hyperkalaemia, thirst, lactic acidosis, hyperglycaemia, hypervolaemia, muscle weakness, syncope, neuropathy, paresthesia, paresis, agitation, confusion, delirium, hallucination, illusion, anxiety, depression, nervousness, anaemia, urinary retention, apnoea, bronchospasm, dyspnea, hypercapnia, hypoventilation, hypoxia, pulmonary congestion, pulmonary hypertension, bronchial obstruction, coughing, emphysema, haemoptysis, pneumothorax, respiratory depression, increased sputum, stridor, increased sweating, erythematous rash, photopsia and abnormal vision.

Investigations: Abnormalities of ECG, T-wave inversion, and an increase in GGT, SGOT, SGPT, alkaline phosphatase, BUN, NPN, or a decreased prothrombin level may be observed.

Drug Abuse and Dependence

The dependence potential of Dexmedetomidine has not been studied in humans. However, since studies in rodents and primates have demonstrated that Dexmedetomidine exhibits pharmacologic actions similar to those of clonidine, it is possible that Dexmedetomidine may produce a clonidine-like withdrawal syndrome upon abrupt discontinuation.

Overdose and treatment

The most common adverse reactions reported in conjunction with overdose in these cases included bradycardia, hypotension, over sedation, somnolence and cardiac arrest. In cases of overdose with clinical symptoms, dexmedetomidine infusion should be reduced or stopped. Expected effects are primarily cardiovascular and should be treated as clinically indicated. At high concentration, hypertension may be more prominent than hypotension. Sinus arrest reversed spontaneously or responded to treatment with atropine and glycopyrrolate. Resuscitation was required in isolated cases of severe overdose resulting in cardiac arrest.

Storage Condition

Store below 30°C. Keep out of reach of children

After dilution, chemical and physical in-use stability has been demonstrated for 24 hours at 30°C and 2°C - 8°C at 4 mcg/mL in all proposed compatible solution except for 20% Mannitol.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to the use are the responsibility of the user and would not normally be longer than 24 hours at 2° to 8°C for all proposed compatible solution except 20% Mannitol, unless dilution was taken place in controlled and validated aseptic conditions.

Dosage form and Packaging Available

Sterile solution for Infusion in 2 mL USP Type I Clear Glass Vial.

Pack size: 5 X 2ml

Product Registration Holder/Importer:

Fresenius Kabi Malaysia Sdn Bhd

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Lot 13, Jalan 51A/225,

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Manufactured by

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